VNRVJIET
VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

Name of the Experiment: Biodata

Experiment No. 1 (a) **Date:** _____

(25PC2EC202) APPLIED PYTHON PROGRAMMING LABORATORY

1.Introduction to Python3

AIM: 1a) Write a program to print your biodata on the screen

SOFTWARE: Python 3.12

Python Code:

```
# Roll No: 25071A7733
print("*****BIO DATA*****")
print("Name: M.Sriharsha Reddy")
print("Branch: EVL")
print("Roll No: 25071A7733")
print("Year: 2nd Year")
```

OUTPUT:

```
========== RESTART: C:\Users\sriha\Documents\Python\Record\programs\25071a77
*****BIO DATA*****
Name: M.Sriharsha Reddy
Branch: EVL
Roll No: 25071A7733
Year: 2nd Year
```

RESULT: The code works perfectly. Learnt to display text on the screen using the print() function.

VNRVJIET
VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

Name of the Experiment: GCD and LCM

Experiment No. 1 (b) **Date:** _____

(25PC2EC202) APPLIED PYTHON PROGRAMMING LABORATORY

AIM: 1b) Write a program to compute GCD and LCM of two numbers by taking input from the user

SOFTWARE: Python 3.12

Python Code:

```
# Roll No: 25071A7733
def gcd(a, b):
    d = a
    e = b
    c = a % b
    while c != 0:
        a = b
        b = c
        c = a % b
    return b

def lcm(a, b):
    return (a * b) // gcd(a, b)

a = int(input("Enter first number: "))
b = int(input("Enter second number: "))

print("GCD of", a, "and", b, "is:", gcd(a, b))
print("LCM of", a, "and", b, "is:", lcm(a, b))
```

OUTPUT:

```
========== RESTART: C:\Users\sriha\Documents\Python\Record\programs\25071a77
Enter first number: 12
Enter second number: 18
GCD of 12 and 18 is: 6
LCM of 12 and 18 is: 36
```

RESULT: GCD and LCM - Learnt the Euclidean algorithm and how LCM relates to GCD.

VNRVJIET
VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

Name of the Experiment: Command Line Arguments

Experiment No. 1 (c) **Date:** _____

(25PC2EC202) APPLIED PYTHON PROGRAMMING LABORATORY

AIM: 1c) Write a program add.py that takes 2 numbers as command line arguments and prints its sum

SOFTWARE: Python 3.12

Python Code:

```
# Roll No: 25071A7733
import sys
if len(sys.argv) != 3:
    print("Usage: python add.py number1 number2")
else:
    n1 = float(sys.argv[1])
    n2 = float(sys.argv[2])
    print("Sum =", n1 + n2)
```

OUTPUT:

```
========== RESTART: C:\Users\sriha\Documents\Python\Record\programs\25071a77
Sum = 42.0
```

RESULT: Learnt to read command line arguments using sys.argv and to convert them to numbers.

VNRVJIET
VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

Name of the Experiment: GCD of Two Numbers

Experiment No. 1 (d) **Date:** _____

(25PC2EC202) APPLIED PYTHON PROGRAMMING LABORATORY

AIM: 1d) Write a program to find the GCD of two numbers by taking input from the user

SOFTWARE: Python 3.12

Python Code:

```
# Roll No: 25071A7733
a=int(input("Enter first number: "))
b=int(input("Enter second number: "))
d=a
e=b
c=a%b
while c!=0:
    a=b
    b=c
    c=a%b
print("GCD of", d, "and", e, "is:", b)
```

OUTPUT:

```
========== RESTART: C:\Users\sriha\Documents\Python\Record\programs\25071a77
Enter first number: 48
Enter second number: 18
GCD of 48 and 18 is: 6
```

RESULT: Learnt to find the GCD of two numbers using the Euclidean algorithm with a while loop.

VNRVJIET
VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

Name of the Experiment: Arithmetic Operations

Experiment No. 1 (e) **Date:** _____

(25PC2EC202) APPLIED PYTHON PROGRAMMING LABORATORY

AIM: 1e) Write a program to perform arithmetic operations on two numbers taken from the user

SOFTWARE: Python 3.12

Python Code:

```
# Roll No: 25071A7733
a= int(input("Enter first number: "))
b= int(input("Enter second number: "))
print("Arithmetic Operations in Python")
print("Addition:", a + b)
print("Subtraction:", a - b)
print("Multiplication:", a * b)
print("Division:", a / b)
print("Floor Division:", a // b)
print("Modulus:", a % b)
print("Exponentiation:", a ** b)
```

OUTPUT:

```
========== RESTART: C:\Users\sriha\Documents\Python\Record\programs\25071a77
Enter first number: 17
Enter second number: 5
Arithmetic Operations in Python
Addition: 22
Subtraction: 12
Multiplication: 85
Division: 3.4
Floor Division: 3
Modulus: 2
Exponentiation: 1419857
```

RESULT: Learnt the arithmetic operators of Python: +, -, *, /, //, % and ** .

VNRVJIET
VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

Name of the Experiment: for Loop over a Sequence

Experiment No. 2 (a) **Date:** _____

(25PC2EC202) APPLIED PYTHON PROGRAMMING LABORATORY

2.Control flow

AIM: 2a) Write programs using for loop that loops over a sequence.

SOFTWARE: Python 3.12

Python Code:

```
# Roll No: 25071A7733
n=10
for num in range(1, n+1):
    print(num)
```

OUTPUT:

```
==================== RESTART: C:\Users\sriha\Documents\Python\Record\programs\25071a77
1
2
3
4
5
6
7
8
9
10
```

RESULT: Learnt the for loop with range() for definite iteration over a sequence.

VNRVJIET
VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

Name of the Experiment: Even or Odd

Experiment No. 2 (b) **Date:** _____

(25PC2EC202) APPLIED PYTHON PROGRAMMING LABORATORY

AIM: 2b) Write a Program for checking whether the given numbers are even or odd

SOFTWARE: Python 3.12

Python Code:

```
# Roll No: 25071A7733
n = int(input("Enter the number of elements: "))

numbers = []

for i in range(n):
    elements = int(input("Enter number: "))
    numbers.append(elements)
    if elements % 2 == 0:
        print(elements, "is even")
    else:
        print(elements, "is odd")
```

OUTPUT:

```
========== RESTART: C:\Users\sriha\Documents\Python\Record\programs\25071a77
Enter the number of elements: 4
Enter number: 12
12 is even
Enter number: 7
7 is odd
Enter number: 20
20 is even
Enter number: 33
33 is odd
```

RESULT: Learnt two-way branching using if-else and the modulus operator.

VNRVJIET
VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

Name of the Experiment: Fibonacci Sequence

Experiment No. 2 (c) **Date:** _____

(25PC2EC202) APPLIED PYTHON PROGRAMMING LABORATORY

AIM: 2c) Write a Program to Print the Fibonacci sequence using for loop.

SOFTWARE: Python 3.12

Python Code:

```
# Roll No: 25071A7733
n = int(input("Enter the number of terms: "))
fibonacci = []
a, b = 0, 1

for i in range(n):
    fibonacci.append(a)
    a, b = b, a + b

print("Fibonacci sequence:", fibonacci)
```

OUTPUT:

```
==================== RESTART: C:\Users\sriha\Documents\Python\Record\programs\25071a77
Enter the number of terms: 10
Fibonacci sequence: [0, 1, 1, 2, 3, 5, 8, 13, 21, 34]
```

RESULT: Learnt to generate the Fibonacci series using a for loop and multiple assignment.

VIGNANA JYOTHI

VNRVJIET
VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

Name of the Experiment: Prime Numbers

Experiment No. 2 (d) **Date:** _____

(25PC2EC202) APPLIED PYTHON PROGRAMMING LABORATORY

AIM: 2d) Write a program to print all prime numbers in a given interval (use break).

SOFTWARE: Python 3.12

Python Code:

```
# Roll No: 25071A7733
start = int(input("Enter starting number: "))
end = int(input("Enter ending number: "))
print("Prime numbers are:")
for num in range(start, end + 1):
    if num < 2:
        continue
    for i in range(2, num):
        if num % i == 0:
            break
    else:
        print(num)
```

OUTPUT:

```
==================== RESTART: C:\Users\sriha\Documents\Python\Record\programs\25071a77
Enter starting number: 10
Enter ending number: 50
Prime numbers are:
11
13
17
19
23
29
31
37
41
43
47
```

RESULT: I learned how to use a for loop with break to check and print prime numbers within a given interval.

VNRVJIET
VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

Name of the Experiment: Even and Odd Lists

Experiment No. 2 (e) **Date:** _____

(25PC2EC202) APPLIED PYTHON PROGRAMMING LABORATORY

AIM: 2e) Write a program to separate the given numbers into a list of even numbers and a list of odd numbers

SOFTWARE: Python 3.12

Python Code:

```
# Roll No: 25071A7733
n = int(input("Enter the number of elements: "))

numbers = []

for i in range(n):
    elements = int(input(f"Enter number {i+1}: "))
    numbers.append(elements)

even_numbers = []
odd_numbers = []

for num in numbers:
    if num % 2 == 0:
        even_numbers.append(num)
    else:
        odd_numbers.append(num)

print("Even numbers:", even_numbers)
print("Odd numbers:", odd_numbers)
```

OUTPUT:

```
==================== RESTART: C:\Users\sriha\Documents\Python\Record\programs\25071a77
Enter the number of elements: 6
Enter number 1: 5
Enter number 2: 12
Enter number 3: 7
Enter number 4: 18
Enter number 5: 21
Enter number 6: 4
Even numbers: [12, 18, 4]
Odd numbers: [5, 7, 21]
```

RESULT: Learnt to store numbers in lists and to separate them into even and odd lists using append().

VIGNANA JYOTHI

VNRVJIET
VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

Name of the Experiment: Positive, Negative or Zero

Experiment No. 2 (f) **Date:** _____

(25PC2EC202) APPLIED PYTHON PROGRAMMING LABORATORY

AIM: 2f) Write a program to check whether each given number is positive, negative or zero

SOFTWARE: Python 3.12

Python Code:

```
# Roll No: 25071A7733
n = int(input("Enter the number of elements: "))

numbers = []
for i in range(n):
    elements = int(input("Enter number: "))
    numbers.append(elements)

for elements in numbers:
    if elements < 0:
        print("negative integer.")
    elif elements > 0:
        print("positive integer.")
    else:
        print("zero.")
```

OUTPUT:

```
==================== RESTART: C:\Users\sriha\Documents\Python\Record\programs\25071a77
Enter the number of elements: 5
Enter number: 15
Enter number: -8
Enter number: 0
Enter number: 23
Enter number: -4
positive integer.
negative integer.
zero.
positive integer.
negative integer.
```

RESULT: Learnt multi-way branching using if-elif-else.

VNRVJIET
VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

Name of the Experiment: Largest Number

Experiment No. 2 (g) **Date:** _____

(25PC2EC202) APPLIED PYTHON PROGRAMMING LABORATORY

AIM: 2g) Write a program to find the largest of four numbers

SOFTWARE: Python 3.12

Python Code:

```
# Roll No: 25071A7733
n = 4
numbers = []
for i in range(n):
    elements = int(input("Enter 4 numbers: "))
    numbers.append(elements)

largest = numbers[0]
for num in numbers:
    if num > largest:
        largest = num
        print("The largest number is:", largest)
```

OUTPUT:

```
========== RESTART: C:\Users\sriha\Documents\Python\Record\programs\25071a77
Enter 4 numbers: 12
Enter 4 numbers: 45
Enter 4 numbers: 7
Enter 4 numbers: 30
The largest number is: 45
```

RESULT: Learnt to find the largest number in a list by comparing each element with the current largest.

VNRVJIET
VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

Name of the Experiment: Leap Year

Experiment No. 2 (h) **Date:** _____

(25PC2EC202) APPLIED PYTHON PROGRAMMING LABORATORY

AIM: 2h) Write a program to check whether the given year is a leap year or not

SOFTWARE: Python 3.12

Python Code:

```
# Roll No: 25071A7733
n=int(input("Enter the year: "))
if (n % 4 == 0 and n % 100 != 0) or (n % 400 == 0):
    print(n, "is a leap year")
else:
    print(n, "is not a leap year")
```

OUTPUT:

```
==================== RESTART: C:\Users\sriha\Documents\Python\Record\programs\25071a77
Enter the year: 2024
2024 is a leap year
```

RESULT: Learnt to combine conditions using logical operators (and, or) to check for a leap year.

VNRVJIET
VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

Name of the Experiment: Sum of N Natural Numbers

Experiment No. 2 (i) **Date:** _____

(25PC2EC202) APPLIED PYTHON PROGRAMMING LABORATORY

AIM: 2i) Write a program to find the sum of first N natural numbers

SOFTWARE: Python 3.12

Python Code:

```
# Roll No: 25071A7733
n = int(input("Enter the number of elements: "))
sum = 0
for i in range(n):
    sum += i + 1
print("The sum of first", n, "natural numbers is:", sum)
```

OUTPUT:

```
==================== RESTART: C:\Users\sriha\Documents\Python\Record\programs\25071a77
Enter the number of elements: 10
The sum of first 10 natural numbers is: 55
```

RESULT: Learnt to accumulate a running sum using a for loop.

VNRVJIET
VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

Name of the Experiment: Multiplication Table

Experiment No. 2 (j) **Date:** _____

(25PC2EC202) APPLIED PYTHON PROGRAMMING LABORATORY

AIM: 2j) Write a program to print the multiplication table of a given number

SOFTWARE: Python 3.12

Python Code:

```
# Roll No: 25071A7733
n = int(input("Enter the number: "))
for i in range(1, 11):
    print(n, "x", i, "=", n * i)
```

OUTPUT:

```
==================== RESTART: C:\Users\sriha\Documents\Python\Record\programs\25071a77
Enter the number: 7
7 x 1 = 7
7 x 2 = 14
7 x 3 = 21
7 x 4 = 28
7 x 5 = 35
7 x 6 = 42
7 x 7 = 49
7 x 8 = 56
7 x 9 = 63
7 x 10 = 70
```

RESULT: Learnt to print a multiplication table using a for loop with range(1, 11).

VNRVJIET
VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

Name of the Experiment: Even Numbers from 1 to 20

Experiment No. 2 (k) **Date:** _____

(25PC2EC202) APPLIED PYTHON PROGRAMMING LABORATORY

AIM: 2k) Write a program to print the even numbers from 1 to 20

SOFTWARE: Python 3.12

Python Code:

```
# Roll No: 25071A7733
n=20
elements = []
for i in range(1, n+1):
    elements.append(i)
for num in elements:
    if num % 2 == 0:
        print(num, "is even")
```

OUTPUT:

```
==================== RESTART: C:\Users\sriha\Documents\Python\Record\programs\25071a77
2 is even
4 is even
6 is even
8 is even
10 is even
12 is even
14 is even
16 is even
18 is even
20 is even
```

RESULT: Learnt to build a list using a for loop and to filter even numbers using the modulus operator.

VNRVJIET
VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

Name of the Experiment: break Statement

Experiment No. 2 (l) **Date:** _____

(25PC2EC202) APPLIED PYTHON PROGRAMMING LABORATORY

AIM: 2l) Write a program to print the numbers from 1 to 10 and terminate the loop at 5 using break

SOFTWARE: Python 3.12

Python Code:

```
# Roll No: 25071A7733
n=10
for num in range(1, n+1):
    print(num)
    if num == 5:
        print("Breaking the loop at 5")
        break
```

OUTPUT:

```
==================== RESTART: C:\Users\sriha\Documents\Python\Record\programs\25071a77
1
2
3
4
5
Breaking the loop at 5
```

RESULT: Learnt to terminate a loop early using the break statement.

VNRVJIET
VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

Name of the Experiment: continue Statement

Experiment No. 2 (m) **Date:** _____

(25PC2EC202) APPLIED PYTHON PROGRAMMING LABORATORY

AIM: 2m) Write a program to print the numbers from 1 to 10 except 5 using continue

SOFTWARE: Python 3.12

Python Code:

```
# Roll No: 25071A7733
n=10
for num in range(1, n+1):
    if num == 5:
        num += 1
        continue
    print(num)
```

OUTPUT:

```
========== RESTART: C:\Users\sriha\Documents\Python\Record\programs\25071a77
1
2
3
4
6
7
8
9
10
```

RESULT: Learnt to skip the current iteration of a loop using the continue statement.

VNRVJIET
VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

Name of the Experiment: Reading Numbers into a List

Experiment No. 2 (n) **Date:** _____

(25PC2EC202) APPLIED PYTHON PROGRAMMING LABORATORY

AIM: 2n) Write a program to read n numbers using a for loop and store them in a list

SOFTWARE: Python 3.12

Python Code:

```
# Roll No: 25071A7733
n = int(input("Enter the number of elements: "))

list = []

for i in range(n):
    elements=int(input("Enter number: "))
    list.append(elements)

print(list )
```

OUTPUT:

```
==================== RESTART: C:\Users\sriha\Documents\Python\Record\programs\25071a77
Enter the number of elements: 5
Enter number: 10
Enter number: 20
Enter number: 30
Enter number: 40
Enter number: 50
[10, 20, 30, 40, 50]
```

RESULT: Learnt to read multiple values using a for loop and to store them in a list using append().

VNRVJIET
VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

Name of the Experiment: Mean, Median and Mode

Experiment No. 3 (a) **Date:** _____

(25PC2EC202) APPLIED PYTHON PROGRAMMING LABORATORY

3) Lists, Dictionary, Tuples, Strings

AIM: 3a) Write a program to find mean, median, mode for the given set of numbers in a list

SOFTWARE: Python 3.12

Python Code:

```
# Roll No: 25071A7733
numbers = [10, 20, 20, 30, 40, 20, 50]
# Mean
mean = sum(numbers) / len(numbers)
# Median
numbers.sort()
n = len(numbers)
if n % 2 == 0:
    median = (numbers[n//2 - 1] + numbers[n//2]) / 2
else:
    median = numbers[n//2]
# Mode
mode = numbers[0]
max_count = 0
for num in numbers:
    count = numbers.count(num)
    if count > max_count:
        max_count = count
        mode = num
print("Mean =", mean)
print("Median =", median)
print("Mode =", mode)
```

OUTPUT:

```
========== RESTART: C:\Users\sriha\Documents\Python\Record\programs\25071a77
Mean = 27.142857142857142
Median = 20
Mode = 20
```

RESULT: I learned how to calculate mean using addition and division, find the median after sorting, and find the mode by counting the frequency of each number.

VNRVJIET
VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

Name of the Experiment: Character Count Dictionary

Experiment No. 3 (b) **Date:** _____

(25PC2EC202) APPLIED PYTHON PROGRAMMING LABORATORY

AIM: 3b) Write a program to count the numbers of characters in the string and store them in a dictionary data structure

SOFTWARE: Python 3.12

Python Code:

```
# Roll No: 25071A7733
text = input("Enter a string: ")
char_count = {}
for ch in text:
    if ch in char_count:
        char_count[ch] += 1
    else:
        char_count[ch] = 1
print("Character count:", char_count)
```

OUTPUT:

```
========== RESTART: C:\Users\sriha\Documents\Python\Record\programs\25071a77
Enter a string: programming
Character count: {'p': 1, 'r': 2, 'o': 1, 'g': 2, 'a': 1, 'm': 2, 'i': 1, 'n': 1}
```

RESULT: I learned how to use a dictionary to store each character and its frequency, and how to use a for loop and conditional statement to count characters.

VNRVJIET
VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

Name of the Experiment: Configuration using Tuple

Experiment No. 3 (c) **Date:** _____

(25PC2EC202) APPLIED PYTHON PROGRAMMING LABORATORY

AIM: 3c) Write a Python program that stores application configuration values using a tuple to ensure immutability. Display all configuration details

SOFTWARE: Python 3.12

Python Code:

```
# Roll No: 25071A7733
config = ("MyApp", "1.0", "localhost", 8080, "Production")
print("Application Configuration")
print("Application Name:", config[0])
print("Version:", config[1])
print("Server:", config[2])
print("Port:", config[3])
print("Environment:", config[4])
```

OUTPUT:

```
========== RESTART: C:\Users\sriha\Documents\Python\Record\programs\25071a77
Application Configuration
Application Name: MyApp
Version: 1.0
Server: localhost
Port: 8080
Environment: Production
```

RESULT: I learned how to use a tuple to store configuration values and access its elements using indexes. Tuples are immutable, so their values cannot be changed after creation.

VIGNANA JYOTHI

VNRVJIET
VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

Name of the Experiment: Palindrome Check

Experiment No. 3 (d) **Date:** _____

(25PC2EC202) APPLIED PYTHON PROGRAMMING LABORATORY

AIM: 3d) Write a program to check whether the given numbers are palindromes or not (use string slicing).

SOFTWARE: Python 3.12

Python Code:

```
# Roll No: 25071A7733
n = int(input("Enter the number of elements: "))
numbers = []

for i in range(n):
    elements = int(input(f"Enter number {i+1}: "))
    numbers.append(elements)

print(f"\nThe numbers you entered are: {numbers}\n")

for number in numbers:
    if str(number) == str(number)[::-1]:
        print(f"{number} is a palindrome\n")
    else:
        print(f"{number} is not a palindrome\n")
```

OUTPUT:

```
========== RESTART: C:\Users\sriha\Documents\Python\Record\programs\25071a77
Enter the number of elements: 4
Enter number 1: 121
Enter number 2: 345
Enter number 3: 1331
Enter number 4: 2024

The numbers you entered are: [121, 345, 1331, 2024]

121 is a palindrome

345 is not a palindrome

1331 is a palindrome

2024 is not a palindrome
```

VNRVJIET
VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

Name of the Experiment: Palindrome Check

Experiment No. 3 (d) **Date:** _____

(25PC2EC202) APPLIED PYTHON PROGRAMMING LABORATORY

RESULT: I learned how to check whether a number is a palindrome by converting it to a string, reversing it using slicing ([::-1]) and comparing it with the original.

VNRVJIET
VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

Name of the Experiment: Simple Calculator

Experiment No. 4 (a) **Date:** _____

(25PC2EC202) APPLIED PYTHON PROGRAMMING LABORATORY

4) Functions

AIM: 4a) Write a function to implement Simple Calculator program

SOFTWARE: Python 3.12

Python Code:

```
# Roll No: 25071A7733
def calculator(a, b, operator):
    if operator == '+':
        return a + b
    elif operator == '-':
        return a - b
    elif operator == '*':
        return a * b
    elif operator == '/':
        return a / b
    else:
        return "Invalid operator"
a = float(input("Enter first number: "))
b = float(input("Enter second number: "))
operator = input("Enter operator (+, -, *, /): ")
result = calculator(a, b, operator)
print("Result =", result)
```

OUTPUT:

```
========== RESTART: C:\Users\sriha\Documents\Python\Record\programs\25071a77
Enter first number: 24
Enter second number: 6
Enter operator (+, -, *, /): /
Result = 4.0
```

RESULT: I learned how to create and call a function to perform arithmetic operations and implement a simple calculator using conditional statements.

VNRVJIET
VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

Name of the Experiment: Factorial by Recursion

Experiment No. 4 (b) **Date:** _____

(25PC2EC202) APPLIED PYTHON PROGRAMMING LABORATORY

AIM: 4b) Write a function to Find the factorial of a number using recursion

SOFTWARE: Python 3.12

Python Code:

```
# Roll No: 25071A7733
def factorial(n):
    if n == 0 or n == 1:
        return 1
    else:
        return n * factorial(n - 1)
num = int(input("Enter a number: "))
print("Factorial =", factorial(num))
```

OUTPUT:

```
==================== RESTART: C:\Users\sriha\Documents\Python\Record\programs\25071a77
Enter a number: 5
Factorial = 120
```

RESULT: I learned how to use recursion, where a function calls itself to solve a problem. Here, the factorial function repeatedly calls itself until it reaches the base condition n == 0 or n == 1.

VNRVJIET
VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

Name of the Experiment: Collatz Function

Experiment No. 4 (c) **Date:** _____

(25PC2EC202) APPLIED PYTHON PROGRAMMING LABORATORY

AIM: 4c) Write a function collatz(x) which does the following: if x is odd, x = 3x + 1; if x is even, then x = x/2. Return the number of steps it takes for x = 1. Display the steps for the numbers 1 to 20

SOFTWARE: Python 3.12

Python Code:

```
# Roll No: 25071A7733
'''find the number of collatz steps for the numbers for numbers 1-20
'''
def collatz_steps(n):
    steps = 0
    while n != 1:
        if n % 2 == 0:   # even
            n = n // 2
        else:            # odd
            n = 3 * n + 1
        steps += 1
    return steps

# Calculate steps for numbers 1 to 20
results = {}
for x in range(1, 21):
    results[x] = collatz_steps(x)

# Display results
print("Number -> Steps")
for num, steps in results.items():
    print(f"{num} -> {steps}")
```

VNRVJIET
VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

Name of the Experiment: Collatz Function

Experiment No. 4 (c) **Date:** _____

(25PC2EC202) APPLIED PYTHON PROGRAMMING LABORATORY

OUTPUT:

```
==================== RESTART: C:\Users\sriha\Documents\Python\Record\programs\25071a77
Number -> Steps
1 -> 0
2 -> 1
3 -> 7
4 -> 2
5 -> 5
6 -> 8
7 -> 16
8 -> 3
9 -> 19
10 -> 6
11 -> 14
12 -> 9
13 -> 9
14 -> 17
15 -> 17
16 -> 4
17 -> 12
18 -> 20
19 -> 20
20 -> 7
```

RESULT: I learned how to write a function that returns the number of Collatz steps taken to reach 1 and how to store the results for 1 to 20 in a dictionary.

VNRVJIET
VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

Name of the Experiment: Normal Distribution

Experiment No. 4 (d) **Date:** _____

(25PC2EC202) APPLIED PYTHON PROGRAMMING LABORATORY

AIM: 4d) Write a function N(x, m, s) = exp(-(x-m)^2/(2s^2)) / (sqrt(2*pi)*s) that computes the Normal distribution

SOFTWARE: Python 3.12

Python Code:

```
# Roll No: 25071A7733
import math
def N(x, m, s):
    return math.exp(-((x - m) ** 2) / (2 * s ** 2)) / (math.sqrt(2 * math.pi) * s)
x = float(input("Enter x: "))
m = float(input("Enter mean (m): "))
s = float(input("Enter standard deviation (s): "))
print("Normal distribution =", N(x, m, s))
```

OUTPUT:

```
==================== RESTART: C:\Users\sriha\Documents\Python\Record\programs\25071a77
Enter x: 1
Enter mean (m): 0
Enter standard deviation (s): 1
Normal distribution = 0.24197072451914337
```

RESULT: I learned how to create a function for the Normal distribution using the mathematical formula, math.exp(), and math.sqrt() in Python.

VNRVJIET
VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

Name of the Experiment: Collatz Sequence Steps

Experiment No. 4 (e) **Date:** _____

(25PC2EC202) APPLIED PYTHON PROGRAMMING LABORATORY

AIM: 4e) Write a function collatz(n) that prints every step of the Collatz sequence of a number and the total number of steps taken to reach 1

SOFTWARE: Python 3.12

Python Code:

```
# Roll No: 25071A7733
def collatz(n):
    steps = 0
    print(f"Starting number: {n}")
    while n != 1:
        if n % 2 == 0:   # even
            n = n // 2
        else:            # odd
            n = 3 * n + 1
        steps += 1
        print(f"Step {steps}: {n}")
    print(f"\nTotal steps taken: {steps}")

x = int(input("Enter a number: "))
collatz(x)
```

VNRVJIET
VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

Name of the Experiment: Collatz Sequence Steps

Experiment No. 4 (e) **Date:** _____

(25PC2EC202) APPLIED PYTHON PROGRAMMING LABORATORY

OUTPUT:

```
==================== RESTART: C:\Users\sriha\Documents\Python\Record\programs\25071a77
Enter a number: 7
Starting number: 7
Step 1: 22
Step 2: 11
Step 3: 34
Step 4: 17
Step 5: 52
Step 6: 26
Step 7: 13
Step 8: 40
Step 9: 20
Step 10: 10
Step 11: 5
Step 12: 16
Step 13: 8
Step 14: 4
Step 15: 2
Step 16: 1

Total steps taken: 16
```

RESULT: I learned how to trace the Collatz sequence step by step using a while loop and count the total number of steps.

VIGNANA JYOTHI

VNRVJIET
VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

Name of the Experiment: Read and Display a File

Experiment No. 5 (a) **Date:** _____

5) Files

AIM: 5a) Write a function to read data from a file and display it on the screen

SOFTWARE: Python 3.12

Python Code:

```
# Roll No: 25071A7733
def read_file(filename):
    f = open(filename, "r")
    data = f.read()
    f.close()
    return data
print(read_file("data.txt"))
```

OUTPUT:

```
========== RESTART: C:\Users\sriha\Documents\Python\Record\programs\25071a77
Python is fun
I love coding
VNRVJIET Hyderabad
```

RESULT: I learned how to open a file in read mode, read its content using read() and display it on the screen using a function.

VNRVJIET
VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

Name of the Experiment: Reverse Each Line of a File

Experiment No. 5 (b) **Date:** _____

(25PC2EC202) APPLIED PYTHON PROGRAMMING LABORATORY

AIM: 5b) Write a program to print each line of a file in reverse order.

SOFTWARE: Python 3.12

Python Code:

```
# Roll No: 25071A7733
f = open("data.txt", "r")
for line in f:
    print(line.strip()[::-1])
f.close()
```

OUTPUT:

```
========== RESTART: C:\Users\sriha\Documents\Python\Record\programs\25071a77
nuf si nohtyP
gnidoc evol I
dabaredyH TEIJVRNV
```

RESULT: I learned how to read a file line by line using a for loop and reverse each line using slicing ([::-1]).

VNRVJIET
VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

Name of the Experiment: Characters, Words and Lines in a File

Experiment No. 5 (c) **Date:** _____

(25PC2EC202) APPLIED PYTHON PROGRAMMING LABORATORY

AIM: 5c) Write a program to compute the number of characters, words and lines in a file.

SOFTWARE: Python 3.12

Python Code:

```
# Roll No: 25071A7733
f = open("data.txt", "r")
text = f.read()
f.close()
lines = text.splitlines()
words = text.split()
print("Number of characters =", len(text))
print("Number of words =", len(words))
print("Number of lines =", len(lines))
```

OUTPUT:

```
==================== RESTART: C:\Users\sriha\Documents\Python\Record\programs\25071a77
Number of characters = 47
Number of words = 8
Number of lines = 3
```

RESULT: I learned how to count the characters, words and lines of a file using len(), split() and the read() method.

VNRVJIET
VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

Name of the Experiment: Character Frequency in a File

Experiment No. 5 (d) **Date:** _____

(25PC2EC202) APPLIED PYTHON PROGRAMMING LABORATORY

AIM: 5d) Write a program to count frequency of characters in a given file.

SOFTWARE: Python 3.12

Python Code:

```
# Roll No: 25071A7733
f = open("data.txt", "r")
text = f.read()
f.close()
freq = {}
for ch in text:
    if ch in freq:
        freq[ch] += 1
    else:
        freq[ch] = 1
for ch in freq:
    print(repr(ch), ":", freq[ch])
```

VIGNANA JYOTHI

VNRVJIET
VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

Name of the Experiment: Character Frequency in a File

Experiment No. 5 (d) **Date:** _____

(25PC2EC202) APPLIED PYTHON PROGRAMMING LABORATORY

OUTPUT:

```
========== RESTART: C:\Users\sriha\Documents\Python\Record\programs\25071a77
'P' : 1
'y' : 2
't' : 1
'h' : 1
'o' : 3
'n' : 3
' ' : 5
'i' : 2
's' : 1
'f' : 1
'u' : 1
'\n' : 3
'I' : 2
'l' : 1
'v' : 1
'e' : 2
'c' : 1
'd' : 3
'g' : 1
'V' : 2
'N' : 1
'R' : 1
'J' : 1
'E' : 1
'T' : 1
'H' : 1
'r' : 1
'a' : 2
'b' : 1
```

RESULT: I learned how to count the frequency of each character in a file by storing the counts in a dictionary.

VNRVJIET
VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

Name of the Experiment: Random Matrix using NumPy

Experiment No. 6 (a) **Date:** _____

(25PC2EC202) APPLIED PYTHON PROGRAMMING LABORATORY

6) The package numpy

AIM: 6a) Creating a matrix of given order m x n containing random numbers in the range 1 to 99999

SOFTWARE: Python 3.12

Python Code:

```
# Roll No: 25071A7733
import numpy as np
m = int(input("Enter number of rows (m): "))
n = int(input("Enter number of columns (n): "))
matrix = np.random.randint(1, 100000, size=(m, n))
print("Matrix of order", m, "x", n)
print(matrix)
```

OUTPUT:

```
==================== RESTART: C:\Users\sriha\Documents\Python\Record\programs\25071a77
Enter number of rows (m): 3
Enter number of columns (n): 4
Matrix of order 3 x 4
[[70347 22506 49155 73343]
 [79193 61500  1390 32180]
 [29309 46362 37641 22583]]
```

RESULT: I learned how to create a matrix of random integers using numpy.random.randint() for a given order m x n.

VNRVJIET
VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

Name of the Experiment: Matrix Operations

Experiment No. 6 (b) Date: _____

(25PC2EC202) APPLIED PYTHON PROGRAMMING LABORATORY

AIM: 6b) Write a program that adds, subtracts and multiplies two matrices. Provide an interface such that, based on the prompt, the function (addition, subtraction, multiplication) should be performed

SOFTWARE: Python 3.12

Python Code:

```
# Roll No: 25071A7733
import numpy as np
def add(a, b):
    return a + b
def subtract(a, b):
    return a - b
def multiply(a, b):
    return np.dot(a, b)
A = np.array([[1, 2], [3, 4]])
B = np.array([[5, 6], [7, 8]])
print("Matrix A =\n", A)
print("Matrix B =\n", B)
choice = input("Enter operation (add / subtract / multiply): ")
if choice == "add":
    print("Result =\n", add(A, B))
elif choice == "subtract":
    print("Result =\n", subtract(A, B))
elif choice == "multiply":
    print("Result =\n", multiply(A, B))
else:
    print("Invalid operation")
```

OUTPUT (add):

```
========== RESTART: C:\Users\sriha\Documents\Python\Record\programs\25071a77
Matrix A =
 [[1 2]
 [3 4]]
Matrix B =
 [[5 6]
 [7 8]]
Enter operation (add / subtract / multiply): add
Result =
 [[ 6  8]
 [10 12]]
```

VIGNANA JYOTHI

VNRVJIET
VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

Name of the Experiment: Matrix Operations

Experiment No. 6 (b) **Date:** _____

(25PC2EC202) APPLIED PYTHON PROGRAMMING LABORATORY

OUTPUT (subtract):

```
==================== RESTART: C:\Users\sriha\Documents\Python\Record\programs\25071a77
Matrix A =
 [[1 2]
 [3 4]]
Matrix B =
 [[5 6]
 [7 8]]
Enter operation (add / subtract / multiply): subtract
Result =
 [[-4 -4]
 [-4 -4]]
```

OUTPUT (multiply):

```
==================== RESTART: C:\Users\sriha\Documents\Python\Record\programs\25071a77
Matrix A =
 [[1 2]
 [3 4]]
Matrix B =
 [[5 6]
 [7 8]]
Enter operation (add / subtract / multiply): multiply
Result =
 [[19 22]
 [43 50]]
```

RESULT: I learned how to add, subtract and multiply matrices using numpy arrays and to choose the operation based on user input.

VNRVJIET
VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

Name of the Experiment: System of Linear Equations

Experiment No. 6 (c) **Date:** _____

(25PC2EC202) APPLIED PYTHON PROGRAMMING LABORATORY

AIM: 6c) Write a program to solve a system of n linear equations in n variables using matrix inverse

SOFTWARE: Python 3.12

Python Code:

```
# Roll No: 25071A7733
import numpy as np
# 2x + y - z = 8
# -3x - y + 2z = -11
# -2x + y + 2z = -3
A = np.array([[2, 1, -1], [-3, -1, 2], [-2, 1, 2]])
b = np.array([8, -11, -3])
A_inv = np.linalg.inv(A)
x = np.dot(A_inv, b)
print("Solution of the equations:")
print("x =", round(x[0], 2))
print("y =", round(x[1], 2))
print("z =", round(x[2], 2))
```

OUTPUT:

```
========== RESTART: C:\Users\sriha\Documents\Python\Record\programs\25071a77
Solution of the equations:
x = 2.0
y = 3.0
z = -1.0
```

RESULT: I learned how to solve a system of linear equations using the matrix inverse, x = A^-1 b, with numpy.linalg.inv().

VNRVJIET
VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

Name of the Experiment: t-test for Equal Means

Experiment No. 7 (a) **Date:** _____

(25PC2EC202) APPLIED PYTHON PROGRAMMING LABORATORY

7) The package scipy and pyplot

AIM: 7a) Finding if two sets of data have the same mean value

SOFTWARE: Python 3.12

Python Code:

```
# Roll No: 25071A7733
from scipy import stats
data1 = [20, 22, 19, 24, 25, 21, 23]
data2 = [30, 28, 32, 29, 31, 27, 33]
t, p = stats.ttest_ind(data1, data2)
print("t-statistic =", t)
print("p-value =", p)
if p > 0.05:
    print("The two sets of data have the same mean")
else:
    print("The two sets of data do not have the same mean")
```

OUTPUT:

```
========== RESTART: C:\Users\sriha\Documents\Python\Record\programs\25071a77
t-statistic = -6.92820323027551
p-value = 1.586602024029914e-05
The two sets of data do not have the same mean
```

RESULT: I learned how to use scipy.stats.ttest_ind() to test whether two sets of data have the same mean using the p-value.

VNRVJIET
VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

Name of the Experiment: Plotting Data from a File

Experiment No. 7 (b) **Date:** _____

(25PC2EC202) APPLIED PYTHON PROGRAMMING LABORATORY

AIM: 7b) Plotting data read from a file

SOFTWARE: Python 3.12

Python Code:

```
# Roll No: 25071A7733
import matplotlib.pyplot as plt
x = []
y = []
f = open("plot_data.txt", "r")
for line in f:
    a, b = line.split()
    x.append(float(a))
    y.append(float(b))
f.close()
plt.plot(x, y, marker="o")
plt.xlabel("x")
plt.ylabel("y")
plt.title("Data read from file")
plt.show()
```

OUTPUT:

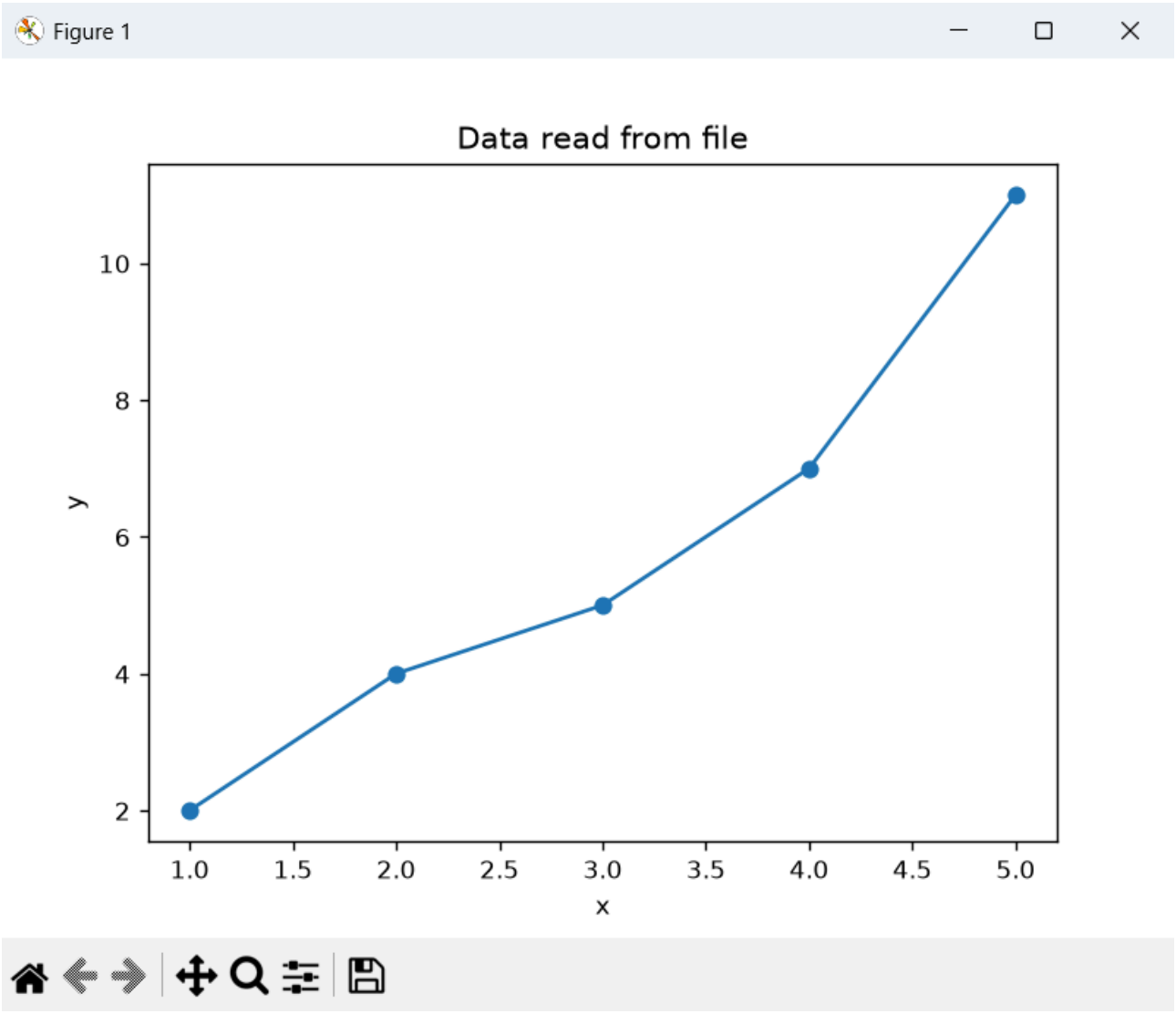
```
========== RESTART: C:\Users\sriha\Documents\Python\Record\programs\25071a77
```

VNRVJIET
VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

Name of the Experiment: Plotting Data from a File

Experiment No. 7 (b) **Date:** _____

(25PC2EC202) APPLIED PYTHON PROGRAMMING LABORATORY



RESULT: I learned how to read x and y values from a file and plot them using matplotlib.pyplot.plot().

VNRVJIET
VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

Name of the Experiment: Curve Fitting using polyfit

Experiment No. 7 (c) **Date:** _____

(25PC2EC202) APPLIED PYTHON PROGRAMMING LABORATORY

AIM: 7c) Fitting a function through a set a data points using polyfit function

SOFTWARE: Python 3.12

Python Code:

```
# Roll No: 25071A7733
import numpy as np
import matplotlib.pyplot as plt
x = np.array([1, 2, 3, 4, 5, 6])
y = np.array([2.2, 4.1, 6.2, 7.9, 10.1, 12.0])
coeff = np.polyfit(x, y, 1)
print("Slope =", coeff[0])
print("Intercept =", coeff[1])
y_fit = np.polyval(coeff, x)
plt.scatter(x, y, label="Data points")
plt.plot(x, y_fit, color="red", label="Fitted line")
plt.legend()
plt.show()
```

OUTPUT:

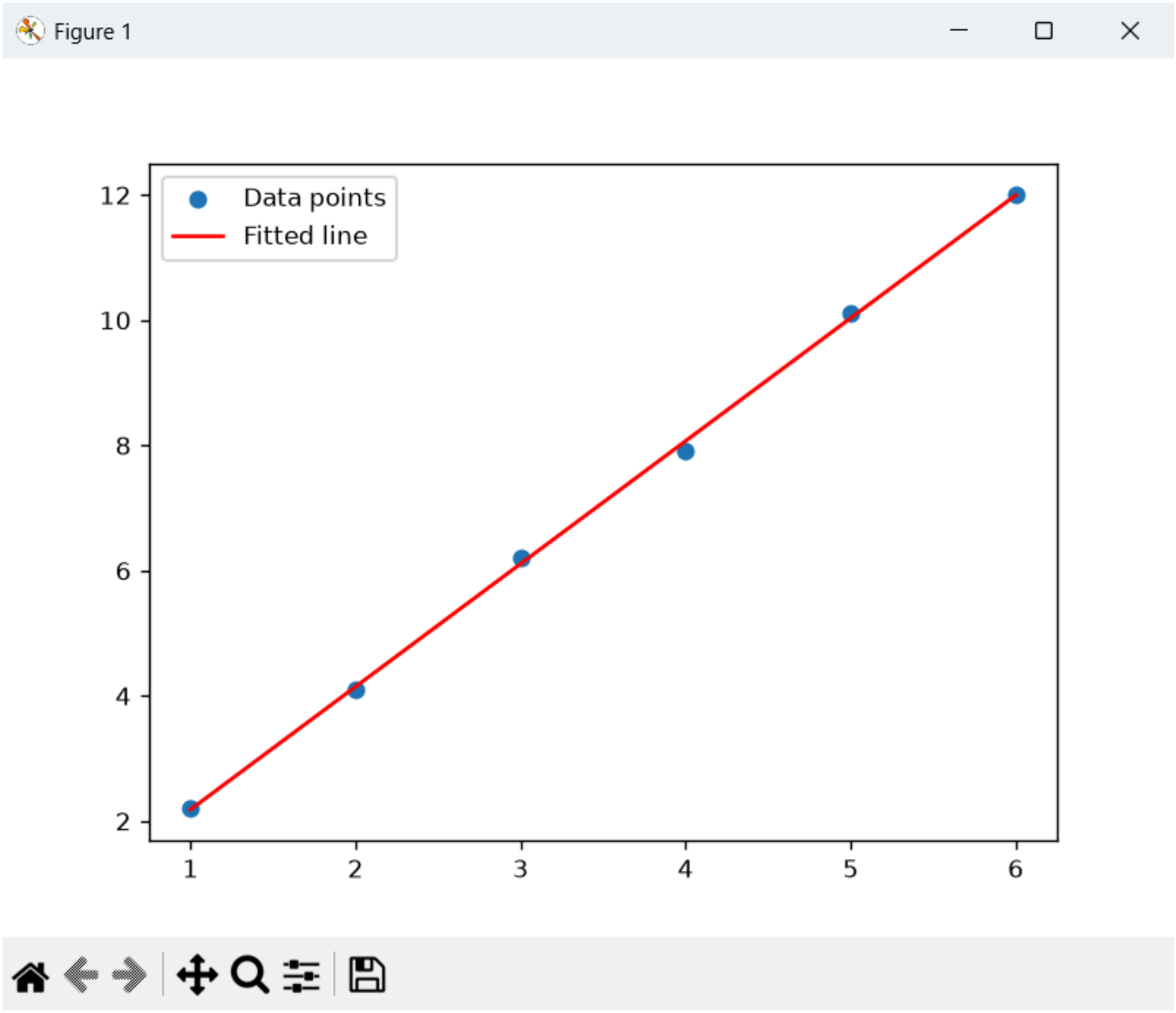
```
========== RESTART: C:\Users\sriha\Documents\Python\Record\programs\25071a77
Slope = 1.9628571428571429
Intercept = 0.21333333333333818
```

VNRVJIET
VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

Name of the Experiment: Curve Fitting using polyfit

Experiment No. 7 (c) **Date:** _____

(25PC2EC202) APPLIED PYTHON PROGRAMMING LABORATORY



RESULT: I learned how to fit a polynomial through data points using numpy.polyfit() and plot the fitted curve with pyplot.

VNRVJIET
VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTE

Name of the Experiment: Histogram of a Data Set

Experiment No. 7 (d) **Date:** _____

(25PC2EC202) APPLIED PYTHON PROGRAMMING LABORATORY

AIM: 7d) Plotting a histogram of a given data set

SOFTWARE: Python 3.12

Python Code:

```
# Roll No: 25071A7733
import matplotlib.pyplot as plt
data = [12, 15, 12, 18, 20, 22, 22, 22, 25, 28, 30, 30, 35, 38, 40, 40, 45]
plt.hist(data, bins=5, edgecolor="black")
plt.xlabel("Values")
plt.ylabel("Frequency")
plt.title("Histogram")
plt.show()
```

OUTPUT:

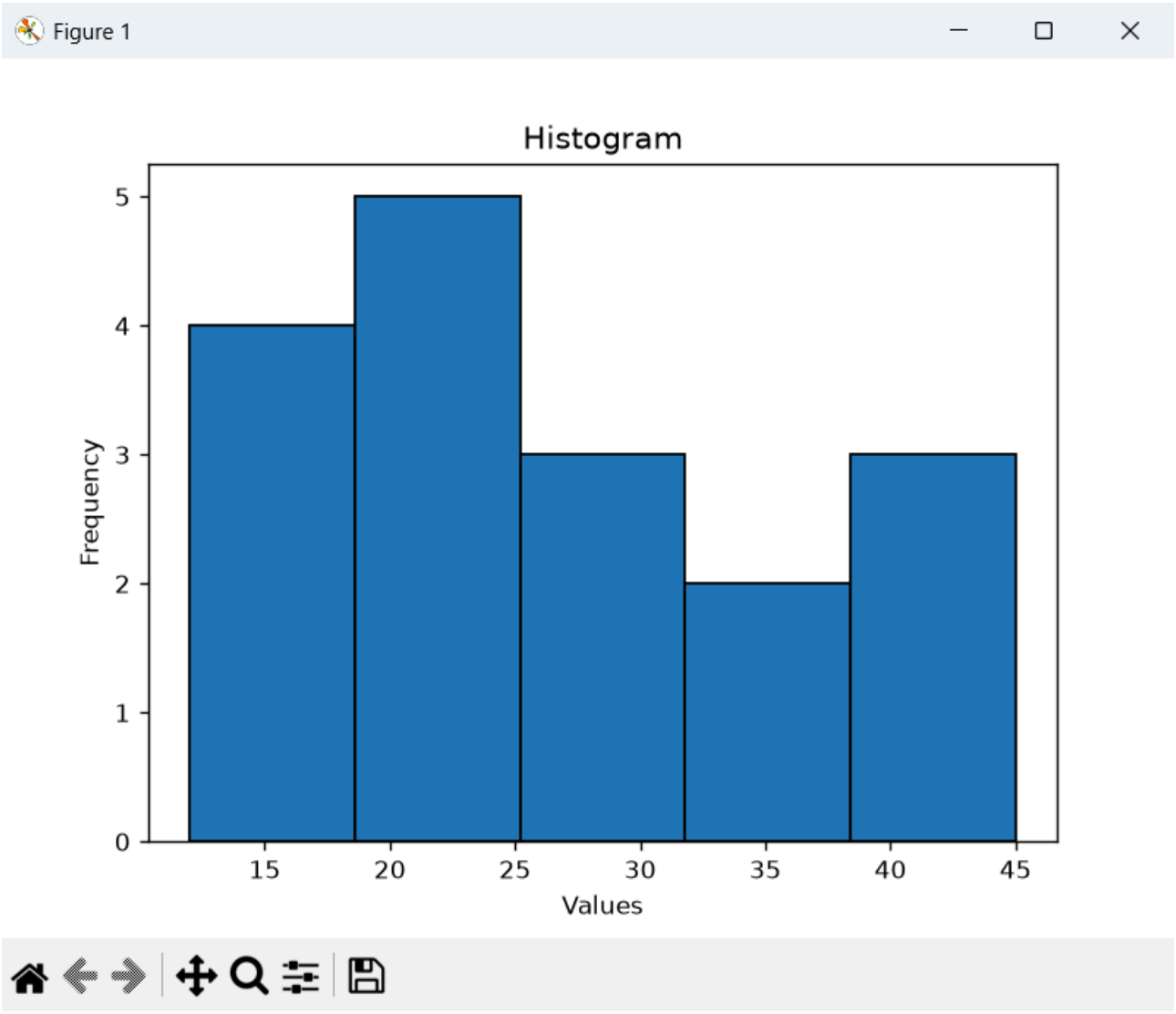
```
========== RESTART: C:\Users\sriha\Documents\Python\Record\programs\25071a77
```

VNRVJIET
VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

Name of the Experiment: Histogram of a Data Set

Experiment No. 7 (d) **Date:** _____

(25PC2EC202) APPLIED PYTHON PROGRAMMING LABORATORY



RESULT: I learned how to plot a histogram of a given data set using pyplot.hist().

VNRVJIET
VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

Name of the Experiment: Blinking LED

Experiment No. 10 (b) **Date:** _____

10) Accessing GPIO pins using Python

AIM: 10b) Write a program to blink an LED connected to one of the GPIO pins

SOFTWARE: Python 3.12

Python Code:

```
# Roll No: 25071A7733
from gpiozero import LED
from time import sleep

# Physical Pin 12 corresponds to BCM GPIO 18
led = LED(18)

try:
    print("Blinking LED on Physical Pin 12... Press Ctrl+C to stop.")
    while True:
        led.on()
        sleep(1)
        led.off()
        sleep(1)

except KeyboardInterrupt:
    print("\nProgram stopped.")
finally:
    led.off()
```

OUTPUT:

```
==================== RESTART: C:\Users\sriha\Documents\Python\Record\programs\25071a77
Blinking LED on Physical Pin 12... Press Ctrl+C to stop.

Program stopped.
```

RESULT: I learned how to use the GPIO Zero library to blink an LED connected to a GPIO pin of the Raspberry Pi using led.on(), led.off() and sleep().

VNRVJIET
VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

Name of the Experiment: Varying LED Brightness

Experiment No. 10 (c) **Date:** _____

(25PC2EC202) APPLIED PYTHON PROGRAMMING LABORATORY

AIM: 10c) Write a program to adjust the brightness of an LED using the in-built PWM

SOFTWARE: Python 3.12

Python Code:

```
# Roll No: 25071A7733
from gpiozero import PWMLED
from time import sleep

# Physical Pin 12 corresponds to BCM GPIO 18 (Hardware PWM)
led = PWMLED(18)

try:
    print("Testing discrete brightness steps...")

    # PWMLED uses a 0.0 to 1.0 floating point scale
    print("Brightness: 25%")
    led.value = 0.25
    sleep(2)

    print("Brightness: 50%")
    led.value = 0.50
    sleep(2)

    print("Brightness: 75%")
    led.value = 0.75
    sleep(2)

    print("Brightness: 100%")
    led.value = 1.0
    sleep(2)

except KeyboardInterrupt:
    print("\nProgram stopped.")
finally:
    led.off()
```

VNRVJIET
VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

Name of the Experiment: Varying LED Brightness

Experiment No. 10 (c) **Date:** _____

(25PC2EC202) APPLIED PYTHON PROGRAMMING LABORATORY

OUTPUT:

```
==================== RESTART: C:\Users\sriha\Documents\Python\Record\programs\25071a77
Testing discrete brightness steps...
Brightness: 25%
Brightness: 50%
Brightness: 75%
Brightness: 100%
```

RESULT: I learned how to vary the brightness of an LED using PWMLED, whose value ranges from 0.0 (off) to 1.0 (full brightness).

VIGNANA JYOTHI

VNRVJIET
VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

Name of the Experiment: Menu Driven Calculator

Experiment No. 11 (a) **Date:** _____

(25PC2EC202) APPLIED PYTHON PROGRAMMING LABORATORY

11) Menu Driven Programs

AIM: 11a) Write a menu driven program to perform addition, subtraction, multiplication and division of two numbers

SOFTWARE: Python 3.12

Python Code:

```
# Roll No: 25071A7733
print("\n===== Menu =====")
print("1. Add two numbers")
print("2. Subtract two numbers")
print("3. Multiply two numbers")
print("4. Divide two numbers")
print("5. Exit")

choice = input("Enter your choice (1-5): ")

if choice == '1':
    num1 = float(input("Enter first number: "))
    num2 = float(input("Enter second number: "))
    print(f"Result: {num1 + num2}")

elif choice == '2':
    num1 = float(input("Enter first number: "))
    num2 = float(input("Enter second number: "))
    print(f"Result: {num1 - num2}")

elif choice == '3':
    num1 = float(input("Enter first number: "))
    num2 = float(input("Enter second number: "))
    print(f"Result: {num1 * num2}")

elif choice == '4':
    num1 = float(input("Enter first number: "))
    num2 = float(input("Enter second number: "))
    if num2 == 0:
        print("Error: Division by zero is not allowed.")
    else:
        print(f"Result: {num1 / num2}")
```

```
elif choice == '5':
    print("Exiting the program. Goodbye!")

else:
    print("Invalid choice. Please select a number from 1 to 5.")
```

OUTPUT:

```
==================== RESTART: C:\Users\sriha\Documents\Python\Record\programs\25071a77

===== Menu =====
1. Add two numbers
2. Subtract two numbers
3. Multiply two numbers
4. Divide two numbers
5. Exit
Enter your choice (1-5): 1
Enter first number: 25
Enter second number: 17
Result: 42.0
```

RESULT: I learned how to build a menu driven program using if-elif-else and f-strings, including a check for division by zero.

VNRVJIET
VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

Name of the Experiment: College Branch Information

Experiment No. 11 (b) **Date:** _____

(25PC2EC202) APPLIED PYTHON PROGRAMMING LABORATORY

AIM: 11b) Write a menu driven program to display the block, floor and room of a branch of the college using lists

SOFTWARE: Python 3.12

Python Code:

```
# Roll No: 25071A7733
# Branches offered at VNRVJIET
branches = [
    "Artificial Intelligence & Data Science",
    "Automobile Engineering",
    "Biotechnology",
    "Civil Engineering",
    "Computer Science & Business Systems",
    "CSE (Cyber Security)",
    "CSE (Data Science)",
    "Computer Science & Engineering (Core)",
    "CSE (AI & ML)",
    "CSE (Internet of Things)",
    "Electronics & Communication Engineering (ECE)",
    "Electrical & Electronics Engineering (EEE)",
    "Electronics & Instrumentation Engineering",
    "Electronics Engineering (VLSI Design & Technology)",
    "Information Technology (IT)",
    "Mechanical Engineering",
    "Robotics & Artificial Intelligence"
]

# Placeholder data for each branch [Block, Floor, Room]
branch_data = [
    ["Block E", "2nd Floor", "Room 201"],   # AI & DS
    ["Block D", "1st Floor", "Room 105"],   # Automobile
    ["Block C", "Ground Floor", "Room 010"],# Biotechnology
    ["Block D", "3rd Floor", "Room 301"],   # Civil
    ["Block E", "2nd Floor", "Room 210"],   # CS & Business Systems
    ["Block F", "3rd Floor", "Room 320"],   # CSE Cyber Security
    ["Block E", "2nd Floor", "Room 215"],   # CSE Data Science
    ["Block E", "4th Floor", "Room 401"],   # CSE Core
    ["Block E", "3rd Floor", "Room 330"],   # CSE AI & ML
    ["Block E", "2nd Floor", "Room 225"],   # CSE IoT
```

```
    ["Block A", "3rd Floor", "Room 340"],   # ECE
    ["Block B", "2nd Floor", "Room 250"],   # EEE
    ["Block B", "1st Floor", "Room 120"],   # EIE
    ["Block C", "3rd Floor", "Room 309"],   # VLSI
    ["Block C", "3rd Floor", "Room 360"],   # IT
    ["Block D", "2nd Floor", "Room 270"],   # Mechanical
    ["Block E", "Ground Floor", "Room 015"] # Robotics & AI
]

# Menu
print("===== Branch Menu =====")
for i, b in enumerate(branches, start=1):
    print(f"{i}. {b}")

choice = int(input("Choose your branch (1-17): "))

if 1 <= choice <= len(branches):
    print(f"\nYou selected {branches[choice-1]}")
else:
    print("Invalid choice. Please select a number from 1 to 17.")
    exit()

# Function to keep looping until Exit
def ask_questions():
    while True:
        print("\n===== Questions Menu =====")
        print("1. What is my Block?")
        print("2. What is my Floor?")
        print("3. What is my Room?")
        print("4. Give me all the information about my branch")
        print("5. Exit")

        q = int(input("Enter your question (1-5): "))

        if q == 1:
            print(f"Block: {branch_data[choice-1][0]}")
        elif q == 2:
            print(f"Floor: {branch_data[choice-1][1]}")
        elif q == 3:
```

```
            print(f"Room: {branch_data[choice-1][2]}")
        elif q == 4:
            print(f"Block: {branch_data[choice-1][0]}")
            print(f"Floor: {branch_data[choice-1][1]}")
            print(f"Room: {branch_data[choice-1][2]}")
        elif q == 5:
            print("Exiting the program. Goodbye!")
            break
        else:
            print("Invalid question choice.")

ask_questions()
```

VIGNANA JYOTHI

VNRVJIET

VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY

AUTONOMOUS INSTITUTE

Name of the Experiment: College Branch Information

Experiment No. 11 (b) **Date:** _____

(25PC2EC202) APPLIED PYTHON PROGRAMMING LABORATORY

OUTPUT:

```
==================== RESTART: C:\Users\sriha\Documents\Python\Record\programs\25071a77
===== Branch Menu =====
1. Artificial Intelligence & Data Science
2. Automobile Engineering
3. Biotechnology
4. Civil Engineering
5. Computer Science & Business Systems
6. CSE (Cyber Security)
7. CSE (Data Science)
8. Computer Science & Engineering (Core)
9. CSE (AI & ML)
10. CSE (Internet of Things)
11. Electronics & Communication Engineering (ECE)
12. Electrical & Electronics Engineering (EEE)
13. Electronics & Instrumentation Engineering
14. Electronics Engineering (VLSI Design & Technology)
15. Information Technology (IT)
16. Mechanical Engineering
17. Robotics & Artificial Intelligence
Choose your branch (1-17): 14

You selected Electronics Engineering (VLSI Design & Technology)

===== Questions Menu =====
1. What is my Block?
2. What is my Floor?
3. What is my Room?
4. Give me all the information about my branch
5. Exit
Enter your question (1-5): 4
Block: Block C
Floor: 3rd Floor
Room: Room 309

===== Questions Menu =====
1. What is my Block?
```

VNRVJIET
VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

Name of the Experiment: College Branch Information

Experiment No. 11 (b) **Date:** _____


```
2. What is my Floor?
3. What is my Room?
4. Give me all the information about my branch
5. Exit
Enter your question (1-5): 5
Exiting the program. Goodbye!
```

RESULT: I learned how to use lists and nested lists to store data, and how to use a while loop to build a menu that repeats until the user chooses to exit.